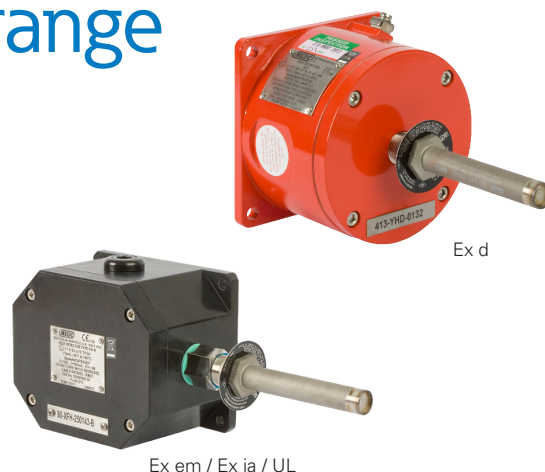


# HD1 heat detector range

Ex d, Ex em & Ex ia



## Overview

The Eaton heat detector has been designed for use in hazardous environments. These units are suitable for fire alarm and/or suppression systems in offshore and onshore applications including paint spray booths, flammable material stores, turbine rooms, extract ductwork and other hazardous areas throughout the oil & gas, petrochemical and process industries.

Comprising a Fenwal rate-compensated detector with all-stainless steel external construction, mounted to either a marine grade alloy enclosure (Exd) or corrosion-free GRP enclosure (Ex ia, Ex em and UL). The contact in the detector CLOSSES at alarm temperature.

To select appropriate temperature setting see specification on reverse.

## Features

- Zone 0, Zone 1 and Zone 2 use
- ATEX / IECEx certified
  - Ex d IIB+H2 T3/T6
  - Ex d IIC T4/T6
  - Ex em IIC T4/T6
  - Ex ia IIC T4/T6 Ga
- UL listed for USA and Canada:
  - Class I, Div 2, Groups A, B, C D
- SIL 2 certified
- IP66 & IP67
- Certified temperature:
  - 20°C to +125°C (Ex d)\*
  - 20°C to +55°C (Ex em / UL)
  - 55°C to +55°C (Ex ia)
- Stainless steel probe (Ex s)
- Detector temperature settings: 60°C to 385°C, (140°F to 725°F)
- Marine grade alloy or GRP enclosure
- Optional stainless steel guard

\*Model dependent



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All specifications, dimensions, weights and tolerances are nominal (typical) and Eaton reserve the right to vary all data without prior notice.  
No liability is accepted for any consequence of use.

## Certifications

|                               |                                                                                                                                                                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ATEX / IECEx EEx d IIB</b> | ATEX cert. no. Baseefa03ATEX0447X<br>IECEX cert. no. IECEX SGS 24.0013X<br>Certified to EN/IEC60079-0, EN/IEC60079-1, EN/IEC60079-31, EN/IEC60079-33<br>Ex II 2 GD, Ex db sb IIB+H2 T6 (T3 @ +125°C)                                                        |
| <b>ATEX / IECEx Ex d IIC</b>  | ATEX cert. no. Baseefa08ATEX0320X<br>IECEX cert. no. IECEX SGS 24.0012X<br>Certified to EN/IEC60079-0, EN/IEC60079-1, EN/IEC60079-31, EN/IEC60079-33<br>Ex II 2 GD, Ex db sb IIC T6 (T4 @ +90°C)*<br>*+90°C not available with addressable module           |
| <b>ATEX / IECEx Ex em</b>     | ATEX cert. no. Baseefa03ATEX0428X<br>IECEX cert. no. IECEX BAS 24.0035X<br>Certified to EN/IEC60079-0, EN/IEC60079-7+A1, EN/IEC60079-33<br>Ex II 2 G, Ex eb sb IIC T6 Gb (-20°C to +55°C), Ex eb em sb IIC T4 Gb (-20°C to +55°C)*<br>*With resistor fitted |
| <b>ATEX / IECEx Ex ia</b>     | ATEX cert. no. Baseefa03ATEX0427<br>IECEX cert. no. IECEX BAS 13.0010<br>Certified to EN/IEC60079-0, EN/IEC60079-11<br>Ex II 1 G, Ex ia IIC T6 Ga (-55°C to +55°C)*<br>*T4 with resistor fitted                                                             |
| <b>UL</b>                     | Listing no. E252920 – versions up to 450°F<br>Listing no. E254077 – versions from 600°F to 725°F<br>UL for USA and Canada, listed to Class 1, Div 2. Groups A, B, C, D                                                                                      |
| <b>SIL</b>                    | SIL2 certified. Cert no. CSA FSP 12007                                                                                                                                                                                                                      |

## Specifications

|                           |                                                                                                                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Material</b>           | Detector: Stainless steel<br>Enclosures: Ex d – LM25 marine grade alloy<br>Ex ia / Ex em / UL – GRP (anti-static)<br>Cover screws: Stainless steel<br>Optional guard: 316 stainless steel                                                 |
| <b>Finish</b>             | Detector: Sand blasted.<br>Enclosures: Ex d painted to customers specification<br>Ex ia / Ex em / UL – Natural black or painted to customer's specification                                                                               |
| <b>Certified temp</b>     | Ex em, UL, Ex d IIC, Ex d IIB: -20°C to +55°C (T6)<br>-20°C to +90°C (T4)<br>Ex d IIB: -20°C to +125°C (T3)<br>-20°C to +55°C (T6)<br>Ex ia: -20°C to +55°C (T4)*<br>-55°C to +55°C (T6)<br>-55°C to +55°C (T4)*<br>*with resistor fitted |
| <b>Weight</b>             | Ex d, 2kg<br>Ex ia / Ex em / UL, 1.1kg                                                                                                                                                                                                    |
| <b>Ingress protection</b> | IP66 & IP67                                                                                                                                                                                                                               |
| <b>Operation</b>          | The detector contact is normally open and CLOSSES at alarm temperature                                                                                                                                                                    |
| <b>Contact rating</b>     | Ex d IIB, Ex em, UL: 125Vac - 5A, 125Vdc - 0.5A, 48Vdc -1A, 24Vdc - 2A<br>Ex d IIC: 24V - 0.1A<br>Ex ia: 30V - 300mA                                                                                                                      |
| <b>Entries</b>            | 2 x M20 (Ex d / Ex em / Ex ia)<br>2 x ½" NPT via adaptors (UL version)                                                                                                                                                                    |
| <b>Terminals</b>          | 6 x 4mm² (BK6)                                                                                                                                                                                                                            |
| <b>Resistor</b>           | Series & EOL resistor (maximum total 2) minimum value (each) 470Ω                                                                                                                                                                         |
| <b>Diodes</b>             | Up to 2 off available in Ex d IIB, Ex d IIC, Ex ia - contact sales office                                                                                                                                                                 |
| <b>Labels</b>             | Optional stainless steel tag and duty labels                                                                                                                                                                                              |

## Ordering requirements

The following code is designed to help in selection of the correct unit. Build up the reference number by inserting the code for each component into the appropriate box

### Model

HD1

### Certification

| Certification                   | Code |
|---------------------------------|------|
| ATEX / IECEx                    | B    |
| UL listed                       | UL   |
| ATEX / IECEx / UL triple listed | AU   |

### Type

| Type     | Code |
|----------|------|
| Ex d IIB | BD*  |
| Ex d IIC | CD*  |
| Ex em    | E†   |
| Ex ia    | I    |
| UL       | U    |

\* Ex d only available in ATEX / IECEx  
† ATEX / IECEx / UL triple listed Ex em version only

### Temp settings

| Temp°F | Code |
|--------|------|
| 140    | 140  |
| 160    | 160  |
| 190    | 190  |
| 210    | 210  |
| 225    | 225  |
| 275    | 275  |
| 325    | 325  |
| 360    | 360  |
| 450    | 450  |
| 500    | 500  |
| 600    | 600  |
| 725    | 725  |

### Options

| Option          | Code |
|-----------------|------|
| None            | N    |
| Duty label      | D*   |
| Tag label       | T*   |
| Guard           | G    |
| Resistor EOL    | E*   |
| Resistor series | S*   |
| Diode(s)        | I*   |
| Blanking plug   | P    |

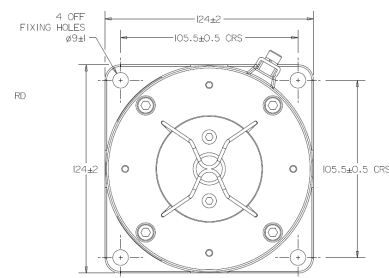
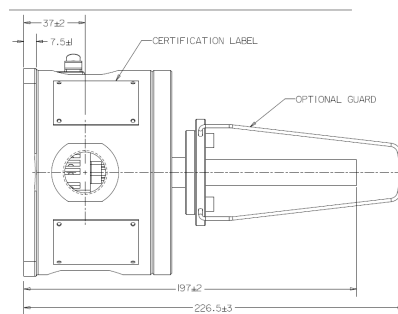
\* Please specify

### Finish

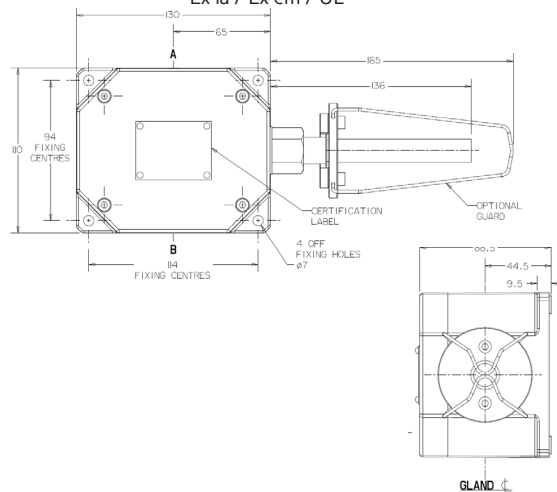
| Finish        | Code |
|---------------|------|
| Natural black | N†   |
| Grey          | G    |
| Red           | R    |
| Blue          | B    |
| Yellow        | Y    |
| Special       | S*   |

\* Please specify  
† Ex em / Ex ia / UL only  
Ex d is painted

## General arrangement drawing (all dimensions in mm)



### Ex ia / Ex em / UL



**Listed temperature settings:** To select appropriate temperature settings, choose detector at 100°F (38°C) above maximum ambient temperature

| Temperature setting |      | Tolerance |      | Colour code of text on probe |
|---------------------|------|-----------|------|------------------------------|
| (°F)                | (°C) | (°F)      | (°C) |                              |
| 140                 | 60   | +7/-8     | ±4   | Black                        |
| 160                 | 71   | +7/-8     | ±4   | Black                        |
| 190                 | 88   | +7/-8     | ±4   | White                        |
| 210                 | 99   | +7/-8     | ±4   | White                        |
| 225                 | 107  | +7/-8     | ±4   | White                        |
| 275                 | 135  | ±10       | ±6   | Blue                         |
| 325                 | 163  | ±10       | ±6   | Red                          |
| 360                 | 182  | ±10       | ±6   | Red                          |
| 450                 | 323  | ±15       | ±8   | Green                        |
| 500                 | 260  | ±15       | ±8   | Orange                       |
| 600                 | 316  | ±20       | ±11  | Orange                       |
| 725                 | 385  | ±25       | ±14  | Orange                       |